



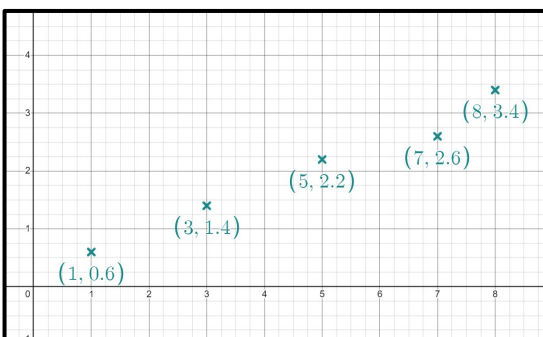
Plotting coordinates generated from a rule using technology

Task A: Plotting coordinates in Desmos

- 1) On your graphing software press the  button if visible.
Change the step on your x axis to 3. Change the step on your y axis to 4
What integer values do your axes range from and to?
- 2) Move your axis until you can see 28 on the y axis. What is the next value visible below 28?
- 3) Press the  button. Describe what this does to your axes.
- 4) Plot the coordinates (2,4) and (2,6). Plot a third coordinate so it is in the middle of these coordinates. What coordinate did you plot?

Task B: Identifying coordinates that do not follow a rule

- 1) Plot each set of coordinates in turn. Identify the coordinate which is the odd one out.
 - a) (0,-4) (4,8) (1,-1) (2,6) (-2,-10)
 - b) (0,7) (4,4) (10,-2) (-1,9) (2,6)
 - c) (8,-1.4) (5,-1.85) (-4, 3.2) (12,-0.8) (-9,-3.95)
 - d) (11,2.9) $(1, \frac{3}{10})$ (-9,-1.1) $(2, 1\frac{1}{10})$ $(-5, -\frac{3}{10})$
- 2) Izzy has plotted the coordinates below.
 - a) Which one do you think she has plotted incorrectly? How do you know?



- b)
 - Recreate Izzy's graph
 - make the incorrect coordinate draggable.
 - Drag the coordinate so it is in line with the rest and on a coordinate where both values are integers.
 - What coordinate have you now plotted?



Task C: Creating shapes in desmos

For each set of coordinates, find the fourth coordinate to make a square

a) $(0,0)$ $(8,-2)$ $(10,6)$

b) $(0,0)$ $(4,5)$ $(-1,9)$

c) $(0,0)$ $(7,-3)$ $(3,7)$

d) $(2,3)$ $(-1,-4)$ $(4,-2)$

e) $(-10,0.5)$ $(-8,1.5)$ $(-9,-1.5)$

2) Plot each set of coordinates to determine the type of quadrilateral they make.

a) $(-12.5,-3)$ $(-12.5, 1.5)$ $(0,-3)$ $(0,1.5)$

c) $(0,9)$ $(3,9)$ $(0,6)$ $(4,5)$

b) $(-8.2,-20)$ $(1.4,-16.3)$ $(-1.2,-16.3)$ $(-5.6,-20)$

d) $(-2,3)$ $(3,3)$ $(2,5)$ $(-1,1)$